Supplementary Step 5 Visualization Report

Overview

Metric	Value
Representative geometry session	PID001 / 2025-11-06_008
Included participant-sessions from Step 5	36
Abstract onset-locked epochs used	539
Concrete onset-locked epochs used	536
Epoch window	-3.5 s to 13.2 s around question onset
Baseline	-2.0 s to 0.0 s
Primary fixed window	7.0 s to 11.0 s
Long-channel montage used in descriptive plots	22 pairs (44 chromophore channels)
Condition mapping for adapted examples	Abstract versus Concrete

Adaptation note. The reference examples were written for a motor task with Tapping, Control, Left, and Right conditions. For the present experiment, those plot families were adapted to the two available task conditions, Abstract and Concrete. All descriptive response-consistency and topographic plots in this report use the Step 5 long-channel montage so that the visualization set stays aligned with the fixed-window ROI analysis.

Rendering note. As in the Step 4 supplementary visualization report, the sensor-geometry figure uses the participant digitization point cloud plus source, detector, channel, and pair geometry rather than an fsaverage brain-surface rendering, because the local `pyvista/fsaverage` stack is not available in this environment.

Step 5 Main Figures Reused

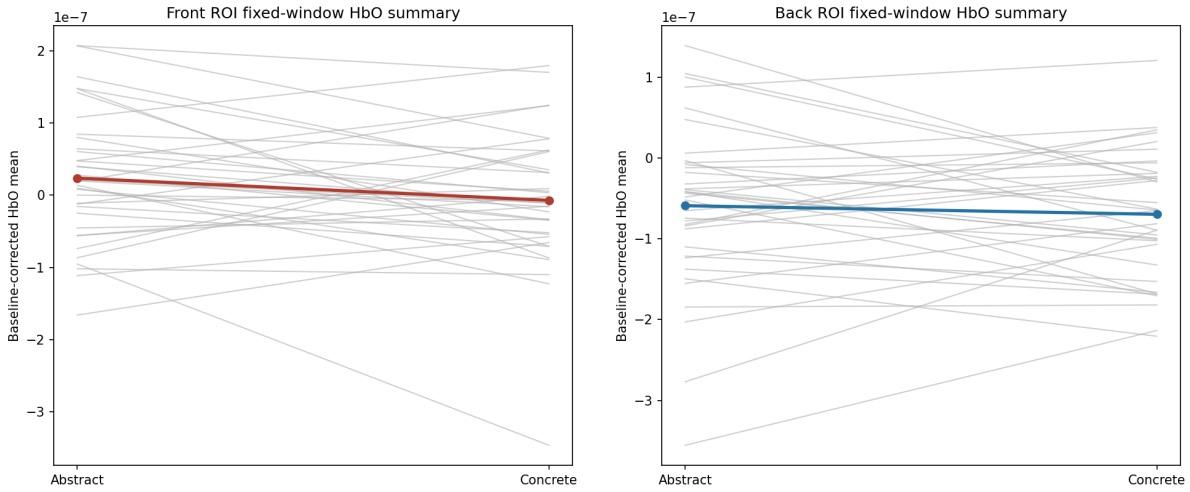

Figure 1: Participant-level fixed-window HbO summaries for the front and back ROIs from the main Step 5 report.

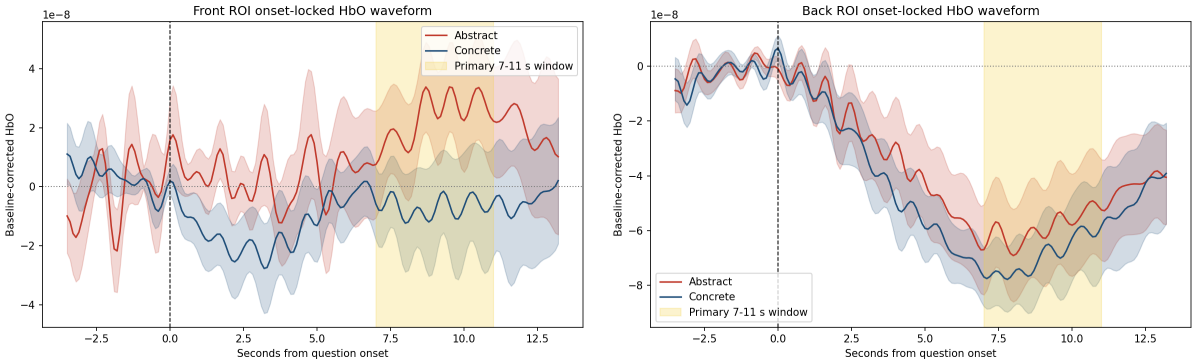

Figure 2: Step 5 onset-locked HbO waveforms with the primary 7–11 s analysis window shaded.

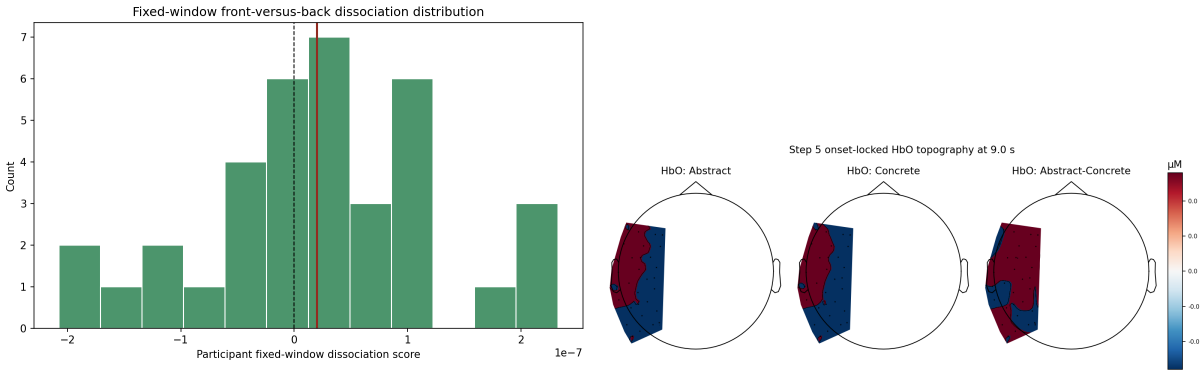

Figure 3: Fixed-window dissociation histogram and the Step 5 single-time HbO topography at 9.0 s.

Sensor Geometry

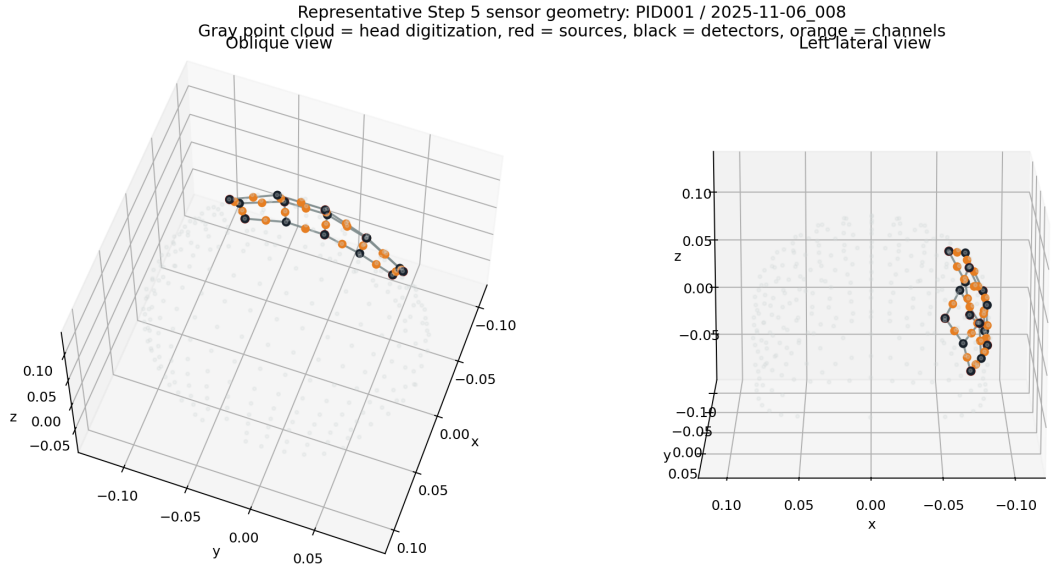

Figure 4: Representative optode, channel, and source-detector-pair geometry for one Step 5 included participant-session. Short pairs are shown with dashed orange lines.

Consistency Across Trials

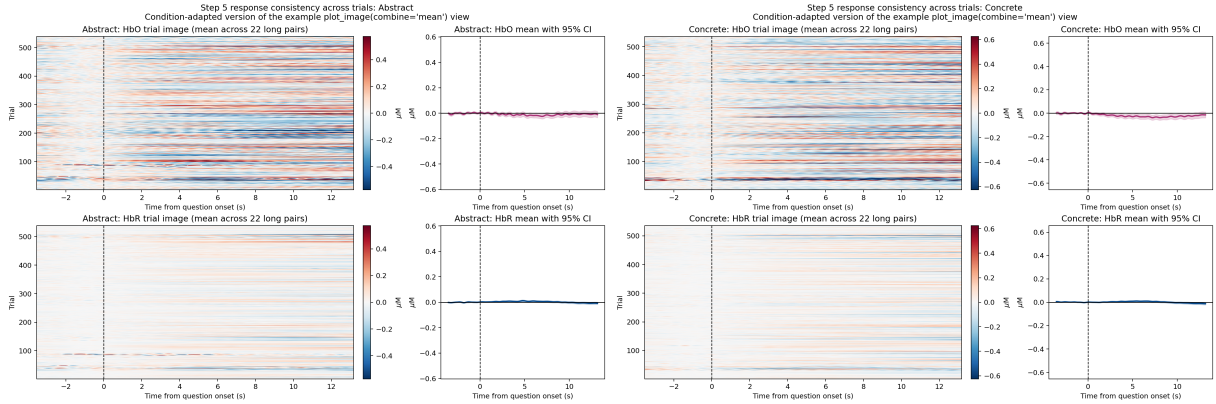

Figure 5: Condition-adapted versions of the trial-consistency image plots. Each panel shows the mean long-channel HbO and HbR response across individual trials for Abstract and Concrete questions separately.

Consistency Across Channels

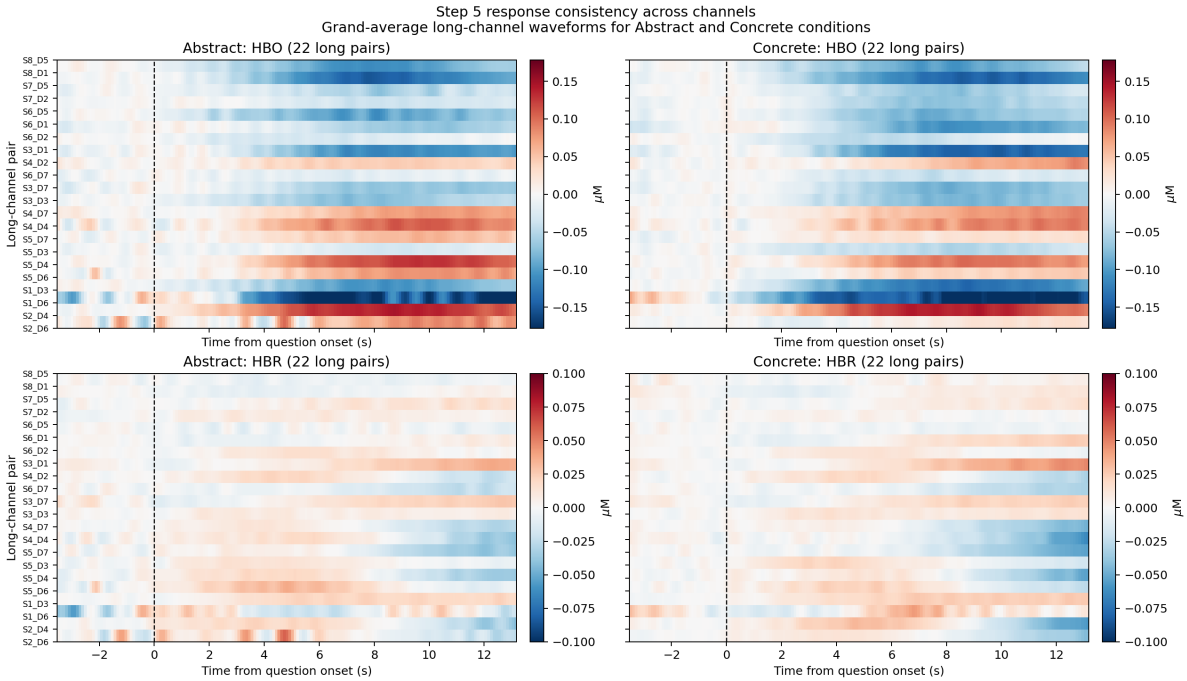

Figure 6: Grand-average long-channel response images across channels for the Abstract and Concrete conditions. Rows correspond to HbO and HbR, and columns correspond to the two conditions.

Standard fNIRS Waveform Summary

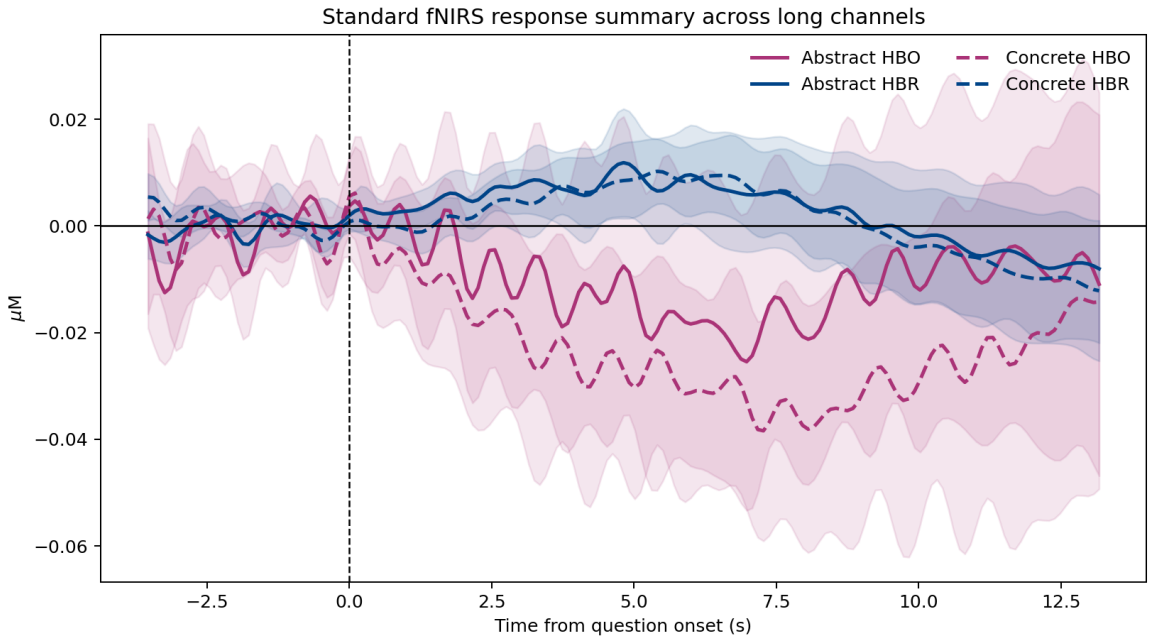

Figure 7: Condition-adapted standard fNIRS waveform plot. Solid lines indicate Abstract, dashed lines indicate Concrete, magenta shows HbO, and blue shows HbR. Shaded bands show 95% confidence intervals across participant-sessions.

Topographic Representation Over Time

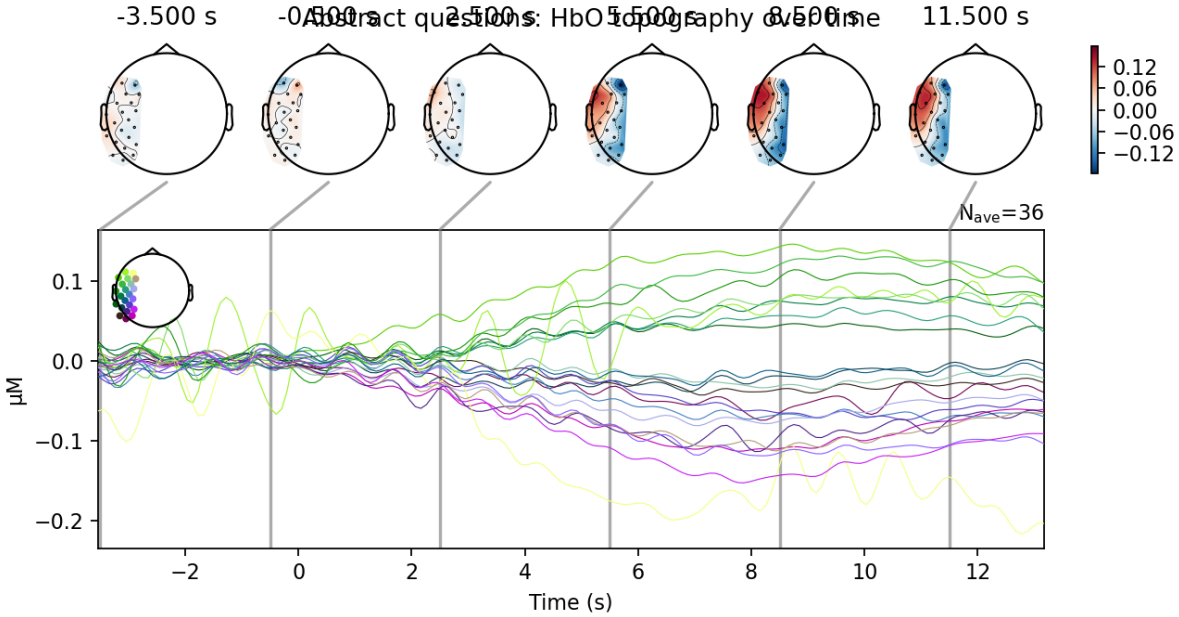

Figure 8: Grand-average HbO topography over time for Abstract questions using the Step 5 included trial set.

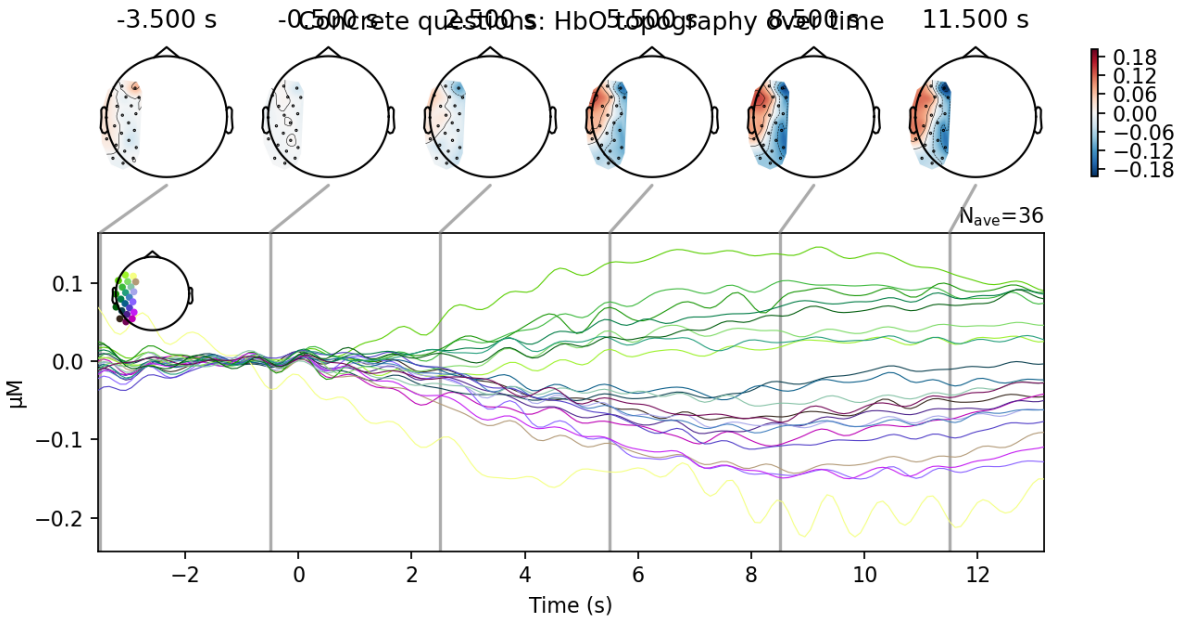

Figure 9: Grand-average HbO topography over time for Concrete questions using the Step 5 included trial set.

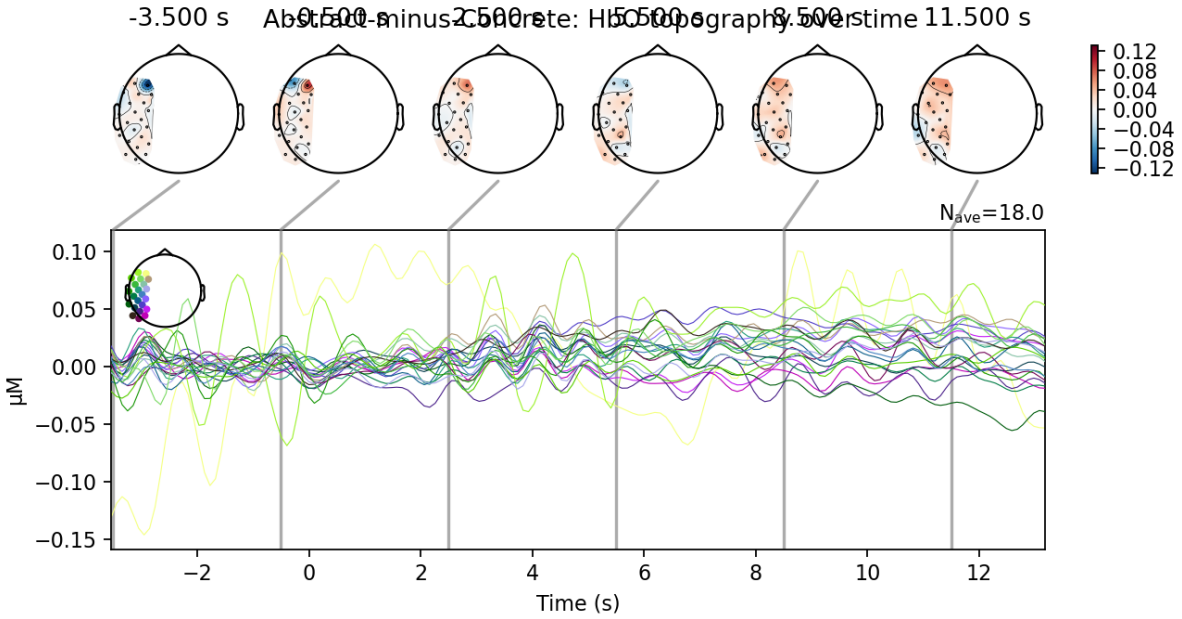

Figure 10: Grand-average Abstract-minus-Concrete HbO topography over time.

Condition Topomap Series

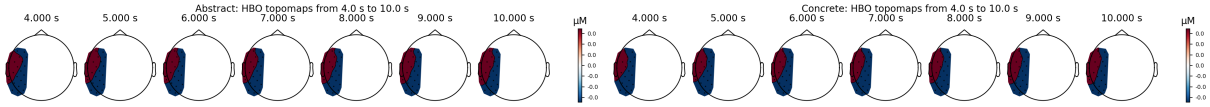

Figure 11: HbO topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

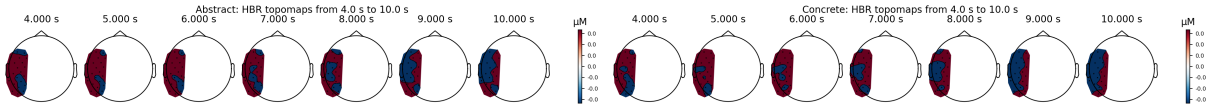

Figure 12: HbR topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

Single-Time Condition Comparison

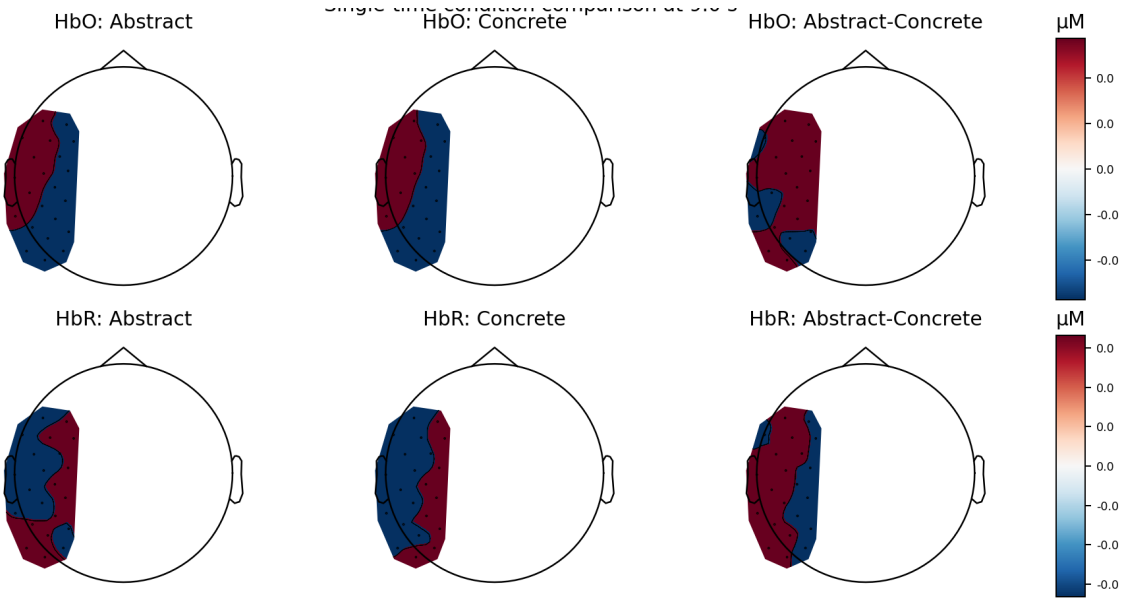

Figure 13: Single-time comparison at 9.0 s for Abstract, Concrete, and Abstract-minus-Concrete topographies for both HbO and HbR.

Channel-Wise Waveform Topography

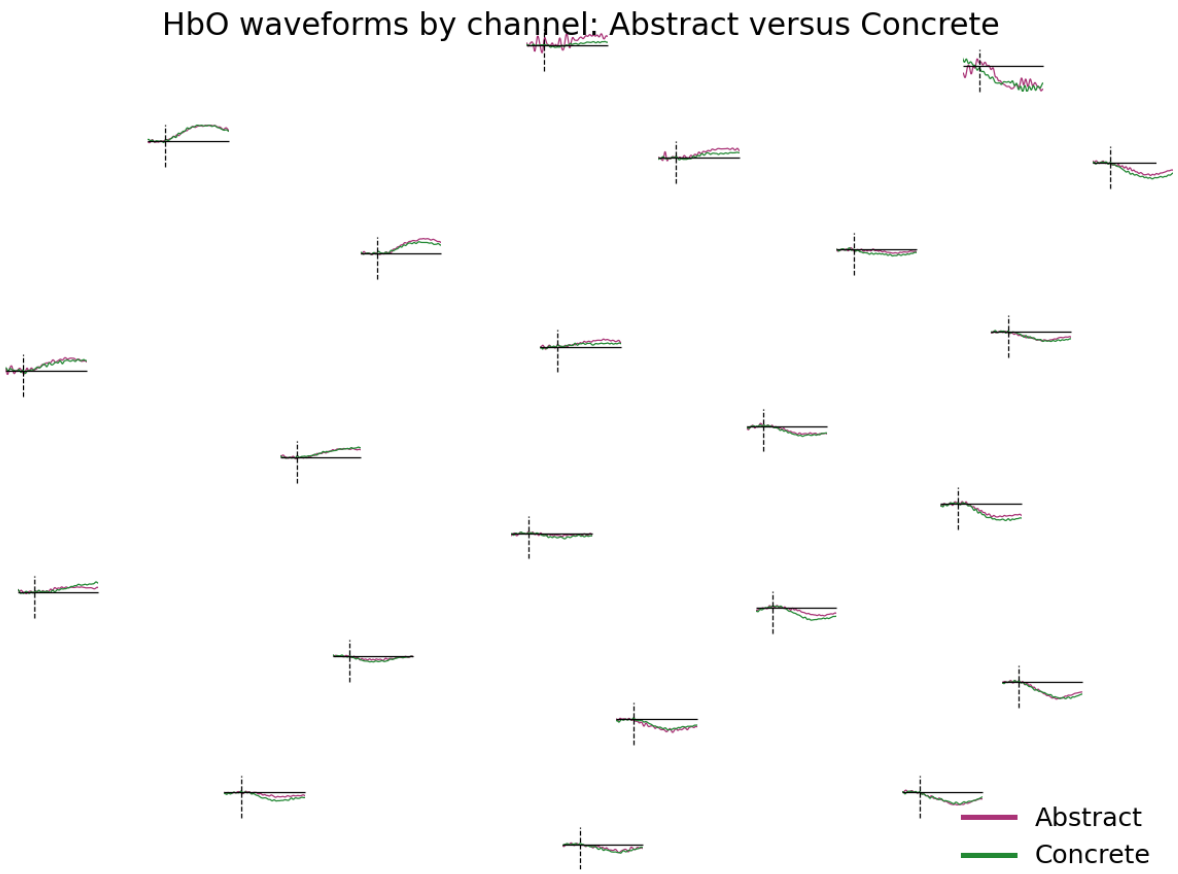

Figure 14: HbO waveforms by long channel, overlaid for Abstract and Concrete, to show which local waveforms drive the descriptive topographic differences.